Rachel Cooray

Analyst, Data and AI, OCTAVE, John Keells Holdings

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I apply AI to real business problems. Analyst, Data and AI at OCTAVE, John Keells Holdings, where I own four AI and analytics use cases for an FMCG business's distributors and design how each one is checked in the field. First Class Computer Science graduate from the University of Westminster, published researcher at IEEE ICAHS 2025, and co-founder of Layer1 Studio.

EXPERIENCE

Analyst, Data and AI, OCTAVE, John Keells Holdings (Colombo, hybrid) Jul 2026 to present

- Own four AI and analytics use cases on the distributor side of an FMCG business: distributor margin incentives, a replenishment system that recommends redistribution volumes, a permanent dispatch plan that minimises lorry costs, and a distributor efficiency model that increases line items per bill.
- Run each end to end: PMO, direct client contact, decks for senior leaders and finance, and coordinating the data science team. Tools: Excel, Databricks, Power BI.
- Design how each use case is checked: pilot and control groups where possible, before and after where not. Start with operational measures such as billing activity, outlet reach and how closely recommendations are followed, then margin. Report results without a control group as an association, not proven impact.
- Join governance calls with area, regional and sales managers. Help prepare answers to finance and leadership challenges. Keep decision logs, process maps and problem statements, and reconcile numbers across dashboards and own the corrections.

Analytics Delivery Associate, OCTAVE, John Keells Holdings (Colombo) Apr 2026 to Jul 2026

Data Science and Analytics Delivery Intern, OCTAVE, John Keells Holdings (Colombo) Oct 2025 to Apr 2026

- Designed KPI-driven distributor incentive structures.
- Ran margin and sales-incentive simulations to recover lost sales in flood-affected outlets, and built Power BI dashboards for management.

Co-founder and Developer, Layer1 Studio (Part-time, remote) Dec 2025 to present

- I co-founded Layer1 Studio and build web and mobile products for small businesses, from shaping a brief to shipping and maintaining it.
- Web and mobile applications in Flutter, React and Node.js, from UI design to backend integration and deployment.
- Clients include gyms, e-commerce, food shops, marketing agencies, a workplace culture consultancy, an artist, an author, sports teams. Interns work on real client projects.

Data Engineering Intern, London Success Academy (Remote, United Kingdom) Sep 2025 to Oct 2025

- Built ETL pipelines on GCP for clickstream and transaction data, scaled processing with PySpark and Parquet partitioning, and produced revenue, daily active user and conversion funnel insights.

Earlier internships at OCTAVE 2022 and 2023

Analytics Delivery Intern, Mar 2023 to May 2023. Data Science Intern, Mar 2022 to Jun 2022.

RESEARCH

PCOS Care: A Machine Learning-Based Web Application for Early Risk Prediction of Polycystic Ovary Syndrome, IEEE ICAHS 2025, Tunisia 2025

- DOI 10.1109/ICAHS66950.2025.11553077. With Malarvizhi Kaniappan Chinnathai and Salma Chahed, University of Westminster.
- An early screening web app with two levels of prediction: one from everyday signs that anyone can answer, and one that adds hormone test results. It shows which factors drove each prediction. It is a screening aid, not a diagnostic tool. Method: Chi-Square and ANOVA feature selection, SMOTE for class balance, and an enhanced Random Forest.
- Third place at the University of Westminster final year project showcase, 2025.
- Extended paper selected for the MSIT Journal special edition, under review.
- Deployed app: pcos-care-web-app.onrender.com. ORCID 0009-0009-3192-2669.

EDUCATION

BSc (Hons) Computer Science, First Class, University of Westminster, United Kingdom Sep 2022 to Jul 2025
Data Science specialism. Started at IIT, Sri Lanka.

UOW/IIT School of Computing and Engineering scholarship, University of Westminster and Informatics Institute of Technology 2024/25

Certificate in Business Accounting, CIMA Aug 2024

St. Bridget's Convent, Colombo, A-Levels: Chemistry, Maths, Physics 2008 to 2022
Head Prefect, 2020.

BUILD LOG

Neurodiversity support chatbot 2024
A chatbot prototype that guides people newly diagnosed with neurodiverse conditions, and their carers, through what comes next. Microsoft EMBRACE hackathon finalist. Generative AI, Conversational design, Prototyping.

Carbon Footprint Tracker
Turns everyday activities into an estimated carbon footprint using activity-based emission factors, with dashboards and suggestions for cutting it. Python, Flutter, Serverpod.

Futsal team recommender
A Flutter app that matches players to teams using collaborative and content-based filtering, with centralised stadium booking. Flutter, Node, Flask, Machine learning.

Fantasy Coin Collector
A 2D dungeon crawler built from scratch in HTML5 Canvas, with its own physics, procedural maps and sound. HTML5 Canvas, JavaScript, Web Audio API.

CSR street-vendor analytics 2022
A field survey of street food vendors whose findings the CSR team adopted. Data collection, Data analytics, Field research.

SKILLS

Languages: Python, SQL, JavaScript, Dart

Data and AI: PySpark, Databricks, Power BI. Explored: scikit-learn, pandas, RAG pipelines

Web and mobile: React, Node.js, Flutter

Cloud and tools: GCP, Git and GitHub, Excel

TEACHING, MENTORING AND OTHER ROLES

Class Teacher, St. Mary's Church, Dehiwala Aug 2022 to Aug 2024

Student Support Worker, Clear Links Support Oct 2024 to Apr 2025

Mentoring interns, Layer1 Studio Dec 2025 to present

Customer-facing roles while studying in the UK
Customer Advisor, UNIQLO, Aug 2025 to Oct 2025. Customer Service Associate, EG On The Move, Jan 2025 to Sep 2025. Retail Associate, TK Maxx, Nov 2024 to Jan 2025.

RECOGNITION

Third place, final year project showcase, University of Westminster, judged with Netcompany and SmartDCC 2025

Hackathon finalist, neurodiversity support chatbot, Microsoft EMBRACE 2024

UOW/IIT School of Computing and Engineering scholarship, University of Westminster and Informatics Institute of Technology 2024/25

New Generation Awards, Women in Management and NYSC 2020